

RAHUL DHIMAN

Mining Technology Lead | FMS & Dispatch, Mine Control, AHS Readiness | 1,000+ Machine Fleet
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PROFESSIONAL SUMMARY

Mining technology leader with 11+ years in open-pit mining and mineral processing in Indonesia and the UAE. Technical owner of the fleet management, dispatch, mine-control and safety platform at Weda Bay Nickel, the world's largest nickel mine: 1,000+ machines, 4 mine sites, 60+ km of haul roads, 7 mining contractors and ~3,000 operators. Wenco Dynamic Dispatch certified; delivered a greenfield FMS from design to go-live (+50% daily production). Leads Autonomous Haulage System (AHS) operational readiness and builds the site's own decision tools. Lean Six Sigma Black Belt; doctoral researcher in AI and autonomous mining.

SELECTED ACHIEVEMENTS

- Lifted daily production from 60,000 to 90,000 t (+50%) with a greenfield Wenco FMS delivered end to end: tower survey, Rajant mesh network, commissioning, go-live and dispatch optimisation.
- Replaced manual dispatch across 1,000+ machines with real-time GPS, speed and payload telemetry; dispatch response time cut from 15–20 min to 2–3 min at 95%+ platform uptime.
- Cut speeding violations ~40% across 3,000+ operators with geofenced speed enforcement, AI driver monitoring and ADAS collision warning; cut unplanned crusher downtime ~50% with automated feed-rate control.
- Built a production simulator on 170,899 measured haul-truck shifts that forecasts trip times, tonnes per route and trucks to roster, and corrected a cycle-time error that overstated forecast production.

CORE COMPETENCIES

FMS & Dispatch: Wenco FMS (Hitachi), Dynamic Dispatch certified; G7 dispatch and safety; dispatch rules and queue configuration; payload management; cycle-time and utilisation analysis; short interval control; FMS vendor trials.

Autonomy & Operational Readiness: AHS readiness assessment; network coverage; high-precision GNSS; geofence and zone design; semi-autonomous and mixed-fleet workflows; remote allocation; operator upskilling; change management.

Mine Control & Safety Technology: 24/7 control-room design; AI driver monitoring (fatigue, seatbelt); ADAS collision warning; geofenced speed enforcement; CCTV; digital radio; ISO 45001; EN ISO 13849-1; AS/NZS 4871 lockout-tagout.

Networks & Operational Technology (OT): Rajant Kinetic Mesh; VHF/UHF, 4G/LTE; fibre, RSTP, Ethernet/IP, Modbus TCP; Trimble GNSS; Siemens S7-1500/1200/300 (TIA Portal); Allen-Bradley ControlLogix (Studio 5000); SCADA/HMI (WinCC, FactoryTalk).

Data & Digital Builds: Power BI (certified), DAX; SQL Server; Python (Flask, FastAPI); React, TypeScript; predictive maintenance; production simulation; live dashboards; drone survey (DJI M300 RTK, Pix4D).

Leadership & Delivery: Technical ownership across 7 contractors and OEM vendors; commissioning and go-live; training for 3,000+ operators; Lean Six Sigma Black Belt (DMAIC, RCA); PMP-trained; Chinese-, Indonesian- and English-speaking teams.

PROFESSIONAL EXPERIENCE

PT ERAMET INDONESIA — WEDA BAY NICKEL (world's largest nickel mine) | Weda, Indonesia

Project FMS Engineer | Technical Adviser, Mining Technology | Oct 2024 – Present

Technical owner of the integrated dispatch, mine-control and safety platform across 4 mine sites, 1,000+ machines, 60+ km of haul roads and 7 mining contractors; project team of ~120.

- Led deployment of the dispatch and safety platform across 1,000+ machines, replacing manual dispatch with real-time GPS, speed and onboard payload telemetry; dispatch response time cut from 15–20 min to 2–3 min.
- Lead Autonomous Haulage System (AHS) operational readiness: network coverage, high-precision positioning, zone design, semi-autonomous workflows and remote allocation logic that a site must complete before autonomy can run.
- Stood up 24/7 mine control rooms with multi-monitor dispatch consoles, CCTV and a live telematics dashboard (FastAPI) that replaced manual daily reports; extended VHF/UHF and 4G coverage across the haul-road network with IT and OT teams, sustaining 95%+ platform uptime.
- Configured geofenced automated speed enforcement by zone and commissioned AI driver monitoring and ADAS forward collision warning; speeding violations down ~40% across 3,000+ operators.
- Built the site digital radio network (repeaters, emergency channel, thousands of radios) tied to verified operator records, and a QR identity system that verifies permit, training, vehicle and job in one scan, replacing paper checks.
- Built an in-house production simulator (Python, SQL Server) on 170,899 measured haul-truck shifts that forecasts trip times, tonnes per route and trucks to roster, and an FMS prototype (React, TypeScript) that became the benchmark for vendor trials.

- Coordinated fleet-wide hardware installation (GNSS, telematics, tablets, cameras, IoT sensors); trained 3,000+ operators and ~50 technical staff; delivered Power BI predictive-maintenance dashboards and DJI M300 RTK drone stockpile surveys ($\pm 2\%$).

STEVIN ROCK LLC | Ras Al Khaimah, UAE

Production Control & Automation Lead Engineer | Feb 2024 – Sep 2024

- Led reliability and safety across PLC-automated crushing operations (~200 personnel in scope); cut unplanned downtime ~50% with real-time monitoring and automatic feed-rate logic that prevents material bridging; weekly emergency stops more than halved.
- Designed and commissioned the crusher control room (video walls, dual-monitor SCADA consoles) and a 58-camera monitoring network, moving operators out of high-noise, high-dust field positions; noise-exposure complaints down ~60%.
- Synchronised quarry truck movements with crusher demand in a continuous plant-to-quarry workflow with the FMS team, reducing downtime on both sides; set and supervised the TPH target for each production period.

Fleet Management System (FMS) Shift Engineer | Jan 2022 – Feb 2024

- As the site's first FMS engineer, delivered a greenfield Wenco FMS from tower survey to go-live; daily production increased from 60,000 to 90,000 t through dynamic dispatch optimisation, onboard payload monitoring and structured operator coaching.
- Led the network build: surveyed tower positions and commissioned solar-powered Rajant towers, radios and repeaters, sustaining a 24-hour mesh network where grid power could not reach.
- Implemented Wenco Dynamic Dispatch: tuned queue, loading and dumping rules shift by shift for Cat 6030/6020B, 993K/994K and Komatsu PC2000 loaders against Terex TR100 and Komatsu HD785 trucks, holding excavator queues under 2 minutes.
- Moved FMS data into SQL and built Power BI dashboards on waiting time, tonnes per hour and payload variance per unit, reviewed hourly with supervisors to revise the plan within the shift; integrated predictive-maintenance alerts with SAP PM.

Automation & Production Engineer / Electrical Engineer | Jan 2016 – Jan 2022

- Owned automation and daily crushing production across a 16-plant crushing complex: delivered 15,000+ lines of Structured Text and Function Block logic (TIA Portal) coordinating 30+ conveyors, crushers and screening plants, sustaining availability above 96%.
- Programmed Allen-Bradley ControlLogix safety interlocks (EN ISO 13849-1 Category 3) and PID feed control holding throughput within $\pm 5\%$; integrated Metso VisioRock OCS-4D optical rock sizing over Modbus TCP with automatic blending correction ($\pm 0.5\%$).
- Commissioned 75+ VFDs and soft starters (ABB, Siemens) on Profibus-DP and Modbus RTU; regenerative braking on decline conveyors cut electrical cost ~10%; planned daily and weekly plant production to customer specification.

EDUCATION

Doctorate in Management, AI & Autonomous Systems in Mining | UCAM, Spain | In progress

M-Tech, Manufacturing Management | BITS Pilani (WILP), India | 2024

University Diploma, Mineral Extractives Studies | University of Derby, United Kingdom | 2024

MBA, Operations & Management | Kalinga University, India | 2015

B-Tech, Electronics & Communication Engineering | Punjab Technical University, India | 2013

CERTIFICATIONS

Wenco FMS Dynamic Dispatch, Wenco International Mining Systems (Hitachi), 2022 | Lean Six Sigma Black Belt, CPD Certified, 2023

Microsoft Power BI Data Analyst Professional Certificate, 2023 | Project Management Professional (PMP) training, 35 contact hours, 2024

McKinsey Forward Program, McKinsey & Company, 2023 | Machine Application Performance, Caterpillar Inc.

ISO 45001 Occupational Health & Safety Management Systems | PLC Automation, Siemens TIA Portal, 2021

Diploma in Quarry Management, Institute of Quarrying, 2018 | Emergency First Aid, Defibrillation and CPR, Highfield, 2024

ADDITIONAL INFORMATION

Target roles: Mining Technology Manager | Mine Control / Dispatch Superintendent | FMS Lead | AHS / Operational Readiness Lead | OT Lead | Mining-tech Product Manager.

Work authorisation: KITAS (Indonesia); Indian national; open to international relocation and FIFO rosters. Member, PMI.

Languages: English (fluent) | Hindi (native) | Bahasa Indonesia (conversational) | French (basic)

Open-source builds: github.com/Lucky0000123 (wbn-fms-simulator, wbn-fms-ui, werci-mobile).